Real-Time Gait Anomaly Detection Using SVM Time Series Classification

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***Abstract*—In this paper, a real-time implementation of Support Vector Machines (SVM) — Real-Time tsSVM Anomaly Detection (RTtsSVM-AD) algorithm is proposed.**

Here, *real-time abnormality detection* is referencing to the ability of the algorithm to detect true gait anomaly occurrence during the swing phase of ongoing step. Anomaly detection is presented with "earliness" measure. For comparative research, eight different human gait deviations were simulated by two healthy volunteers. Corresponding gyroscope angular velocities, from the sensor placed on the forefoot, were recorded. F1 score, true positive rate (TPR), false positive rate (FPR) and "earliness" values were estimated and analyzed.

Real-time classification results, where classification is performed during the ongoing step, are different from regular classification results, where classification is performed after the full step. Thus, they can not be compared directly.

Achieved results prove the concept, that it is possible to detect anomalies in real-time during the swing phase of a step with RTtsSVM-AD algorithm. Best F1 scores for first person's gait recordings were 57%, 53% and 52% for Steppage, Parkinsonian and Ataxic gait types respectively. For the second person's gait recordings, best F1 scores were 65%, 58% and 50% for Slap, Steppage and Hemiplegic gait types respectively.

RTtsSVM-AD algorithm would be developed further and could be used as a base method for comparison with other algorithms.

***Index Terms*—Real-time, Gait analysis, Anomaly detection, Machine learning, Wearable sensors.**

I. Introduction

According to World Health Organisation (WHO) report about one billion persons are affected by neurological disorders worldwide [1]. Neurological diseases ranging from migraine to stroke, and Alzheimer are the leading causes of Disability Adjusted Life Years (DALY) loss [2]. For instance, there is a substantial risk of falling for patients with gait impairments from neurological diseases [3]. It is especially true for patients suffering from neuromuscular diseases, because high variability and deviations from the optimal gait pattern can be seen in their gait [4]. Therefore, it is difficult to analyze patients' gait patterns in real-time. Certain diseases cause abrupt gait changes, such as freezing episodes of Parkinson Disease (PD) [5]. Other diseases, like Multiple Sclerosis (MS) may contain long duration of relapse episodes, with individual impact and slow progression [6]. This means, that neurodegenerative diseases can affect gait quality and change its locomotion cycle to abnormal. The gait of a person can be described by a set of parameters such as: step length, duration of individual step phases, muscle force, etc. [7]. Wearable motion sensors, containing multidimensional Inertial Measurement Units (IMUs), are the most widely used gait assessment devices in recent years for supporting daily activities [8]. IMUs are also used for gait assessment of patients with neurological diseases. For example, data collected from IMUs is used to detect initial and final contact events of the gait cycle of different persons - healthy, stroke and with other neurological disorders, and select best algorithms and sensor placements for correct classification between them [9]. Also, IMUs can be used to detect activities of daily life, fall events and their directions [10], to determine gait parameters and for identification of persons [11], [12], [13]. Finally, IMUs can be used to discover environment dependent differences in gait, which will help with context-aware decisions [14].

It is shown that Functional Electrical Stimulation (FES) can be used to assist walking and help with fall prevention [15] as well as for generic gait improvements [16]. Long term gait deviation analysis and efficient run-time control of FES devices requires automated real-time recognition of abnormal steps or other gait deviations.

Existing real-time algorithms are used for following: identification by gait [17]; for detecting of gait events like heel-strike and toe-off for elderly healthy subjects; stroke patients and patients with Parkinson disease [18], as well as with other impairments [19], [20]; haptic biofeedback devices are implemented using inertial measurement units (IMUs), to correct toe-in or toe-out during walking in real-time [21].

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Notably, none of the previous studies explored real-time anomaly detection in ongoing step, nor base methods are proposed for real-time in step anomaly detection.

According to the previous paper [22], it has been concluded that Support Vector Machines (SVM) based methods are most widely used ones for automated gait analysis, followed by Convolutional Neural Networks (CNN). The benefits of SVM include capability to operate with relatively small datasets and high computational efficiency [10], [23]. However, there is limited research conducted of analysing how well machine learning methods, particularly SVM, performs in detecting realistic gait deviations in real-time during the ongoing step, caused by actual neurological diseases. In our previous paper it was shown, that *tslearn SVM* algorithm implementation [24] can classify normal steps from abnormal and achieve high F1 scores [25] for regular offline full dataset classification. However it can not be used directly for real-time abnormality detection. Thus additional framework should be developed to be able to use *tslearn SVM* in real-time application.

In the view of above discussion, in this paper we provide following contributions:

- Collecting volunteers' walking motion data, with simulated gait deviations, using an industry-standard wearable motion sensor (Shimmer3 IMUs) according to the clinical trial protocol approved by Estonian National Institute for Health Development, permission No.818.
- Proposing novel real-time anomaly detection algorithm, Real-Time tsSVM Anomaly Detection (RTtsSVM-AD) algorithm, which can detect gait abnormalities in real-time during the swing phase of an ongoing step.

This paper consist of five sections: after the introductory state-of-the-art overview, in section II data acquisition and gait types are described, as well as metrics used for analysis; in section III novel RTtsSVM-AD algorithm is described, including data preparation; parameters of the algorithm, results and discussion are presented in the section IV and paper is concluded in section V.

II. Methods

A. Data Acquisition

Eight types of the human gait abnormalities were selected, which were simulated by 2 healthy persons of different gender, both 23 years old, while walking in straight line (Table I).

Collected data was labeled step wise, thus all steps were annotated as *normal* or *abnormal*.

TABLE I: *Labeled gait recordings, collected for this study.*

Person №	Ataxic	Diplegic	Hemiplegic	Hyperkinetic	Parkinsonian	Slap	Steppage	Trendelenburg	Total
p1	1	5	2	2	2	3	5	2	21
p2	4	6	5	4	4	5	9	4	41
Total	5	11	7	6	6	8	14	6	62

Simulations were recreating actual patients' gait recordings on videos and instructions from a professional physiotherapist. Detailed data collection process was described in previous paper [25]. Most frequent gait abnormalities were chosen: Ataxic, Diplegic, Hemiplegic, Hyperkinetic, Parkinonian, Slap, Steppage and Trendelenburg (lurch) gait types [26].

B. Evaluation metrics

For evaluation, several calculated parameters were used, which include F1 score, true positive rate (TPR), false positive rate (FPR) and *earliness*. *Earliness* in this paper is defined as – time between the beginning of a step and moment in time when step was correctly classified as abnormal. The minimal achievable earliness naturally depends on the gait deviation type. Such measure has been introduced, because concrete moment when anomaly is starting to occur can fluctuate, depending on a gait type.

III. Proposed algorithm

In this paper new algorithm is presented, which is able to detect anomalies during the ongoing step of human gait. Such, in-step SVM based anomaly detector was named Real-Time tsSVM Anomaly Detection Algorithm (RTtsSVM-AD). Differences from regular *tslearn SVM* [24] is added framework, to be able to classify incoming signal of ongoing step, which includes automatic hyperparameter optimization and real-time classification.

Hypothesis of RTtsSVM-AD algorithm is following: if by cumulatively replacing the model step chunks with new chunks from ongoing step, collected from IMU placed on a forefoot, the algorithm can classify ongoing step as normal or abnormal, then it is possible to detect abnormality during the swing phase of this step in real-time.

A. Data preparation

Each person datasets are assessed separately. Data is prepared by taking all accessible recordings for current gait type, except for one gait recording, which would be used as validation dataset for real-time step anomaly detection estimation. All other recordings are combined into one dataset and divided into training and test datasets with ratio of 70%:30% correspondingly. Then each step from these datasets is resampled to constant length and normalized.

B. Algorithm description

Algorithm has two phases: training and real-time classification phases. In the proposed implementation synchronous time series data is used to train classifier. For every recording gyroscope vector magnitude is calculated, using the L_2 norm (1) from gyroscope angular velocities around sensor axes.

$$\mathbf{Mag} = \sqrt{\mathbf{X}^2 + \mathbf{Y}^2 + \mathbf{Z}^2} \tag{1}$$

where $\mathbf{X}, \mathbf{Y}$ and $\mathbf{Z}$ are vectors of gyroscope angular velocities around sensor axes, $\mathbf{X} = [x_0, x_1, \ldots, x_i, \ldots, x_n]^T$, $\mathbf{Y} = [y_0, y_1, \ldots, y_i, \ldots, y_n]^T$ and $\mathbf{Z} = [z_0, z_1, \ldots, z_i, \ldots, z_n]^T$,

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sample index $n \in \mathbb{N}$. The $\mathbf{Mag}(X, Y, Z)$ is the gyroscope vector magnitude.

RTtsSVM-AD algorithm training phase is following:

- First is training of the classifiers with particular value for hyper-parameter *gamma* (γ) on training dataset;
- Second step is to estimate classification quality of these classifiers on test dataset using F1 score;
- Best classifiers and corresponding model steps for these classifiers are used in real-time classification of validation dataset.

Hyper-parameter γ is the main parameter of the proposed algorithm, which is responsible for smoothing of the resulting metric for soft-DTW [24]. Quality of classification for this particular γ value is estimated with test dataset. For this hyper-parameter optimization from predetermined options is performed and it stops:

- If F1 score on test dataset for given γ value is 100%;
- If for three different γ values F1 score is the same;
- If all given γ values have been tested.

After training phase, real-time classification performance is estimated, which is described in more details in next subsection. For this best classifiers and model steps are used from the training phase. Several classifiers could be chosen, if they have similar performance, which should increase robustness of the algorithm. Model step is calculated as normal step ensemble average from test dataset, where normal steps that have been classified correctly were chosen. For correct operation of algorithm in real-time phase, data used for model step calculation is not resampled and not normalized.

C. Anomaly prediction

Real-time classification is performed in online-fashion. Validation dataset is fed to the algorithm as streaming data, which arrives as a series of packages or chunks. Step start and step end events are detected when incoming streaming gyroscope vector magnitude is exceeding certain value. The size of each chunk is selected small enough for real-time operation (for reasonable latency of anomaly alarming) and large enough for more efficient processing and data transferring through communication channels. In the current implementation the amount of collected data in one chunk is M samples for each gyroscope vector. Classification is performed on whole classification step. For this incoming chunks are replacing corresponding chunks in model step (Fig. 1). Then this classification step is resampled and normalized. After that probability score is obtained, by classification of classification step by classifier or classifiers. This scores are collected to score buffer. Resulting anomaly detection $Score$ is value (2), which will be compared with the selected *threshold*. If *threshold* has been exceeded, alarm is triggered (3), which is finalizing anomaly detection.

$$Score = \frac{\sum Scores}{N_{cl}}, \tag{2}$$

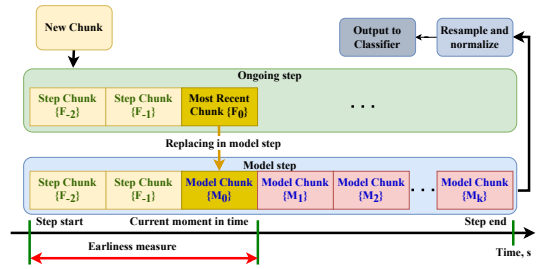

Fig. 1: Replacing chunks from incoming data to model step and resulting step for classification. If anomaly is detected in the current moment of time, earliness is time, shown on figure.

where N_{cl} is the number of classifiers.

$$Alarm = \begin{cases} 1, & \text{if } S > threshold \\ 0, & \text{if } S \leq threshold \end{cases} \tag{3}$$

If alarm is triggered, then earliness is time duration from step start to current moment in time (Fig. 1).

After the end of the current step, model step is restored to original state and real-time classification is repeated for the next step.

IV. Results and Discussion

Results for the RTtsSVM-AD algorithm are presented in current section.

Parameters used in this work are following: sample rate is set to 256 $Samples/s$, one chunk is $M = 12$ samples long which is approximately 43 ms. Predefined γ values are in the range from 100 to 1000 with increase of 100 and in the range from 5 to 100 with increase of 10. Gyroscope angular value for step detection is 200 °/s.

TABLE II: Best thresholds across gait types

Gait type	p1	p2	avg.
Ataxic	0,7	-	0,7
Diplegic	0,6	0,6	0,6
Hemiplegic	0,9	0,8	0,85
Hyperkinetic	0,8	0,8	0,8
Parkinsonian	0,9	0,5	0,7
Slap	0,8	0,4	0,6
Steppage	0,8	0,7	0,75
Trendelenburg	0,9	0,1	0,5

Aforementioned threshold is chosen after real-time phase, where all scores are obtained, because in real world use case best threshold would be adjusted individually for each person (Table. II). Lets observe F1 scores for different gait types. In terms of F1 score (Fig 2) it could be observed, that algorithm is able to detect abnormalities in real-time for most of the gait types. Best results for first person was for Steppage, Parkinsonian and Ataxic gait types, with F1 scores of 57%, 53% and 52% respectively.

For second person best results were for Slap, Steppage and Hemiplegic gait types with F1 scores of 65%, 58% and 50% respectively. Acceptable results were seen as well for

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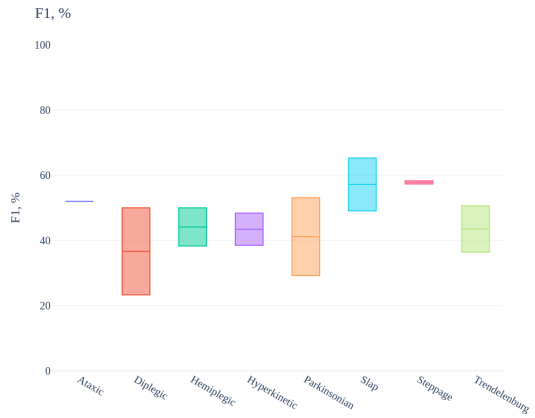

Fig. 2: Distribution of F1 scores for all persons for different gait types. On y-axis is F1 score in percents, on x-axis is different anomalies. Most frequent results are in the boxes, and outliers are shown by whiskers and dots.

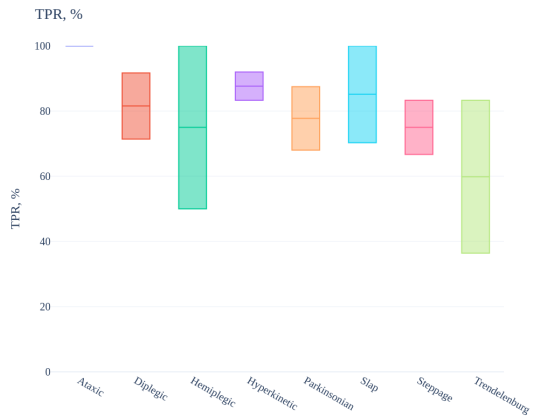

Fig. 3: Distribution of True Positive Rate for all persons for different gait types. On y-axis is True Positive Rate in percents, on x-axis is different anomalies. Most frequent results are in the boxes, and outliers are shown by whiskers and dots.

Trendelenburg and Diplegic gait types with F1 scores of 51% and 50% respectively for first person. Steppage gait type have noticeable difference from the normal step, which makes it easier to classify for two class classifier. Worst performing gait types were Hyperkinetic, Hemiplegic and Slap gait types for first person with F1 scores of 48%, 38% and 49% respectively. For second person it was Diplegic, Hyperkinetic, Parkinsonian and Trendelenburg gait types with F1 score of 23%, 38%, 29% and 36% respectively. Hemiplegic gait type is different from normal gait by swing of a leg to the side, which on magnitude graph does not have major differences from normal step. Average F1 scores shows, that algorithm is able to detect abnormalities in real-time for most gait types, presented in this paper.

For Slap and Parkinsonian gait recordings of first person and Trendelenburg, Hemiplegic, Steppage and Diplegic gait

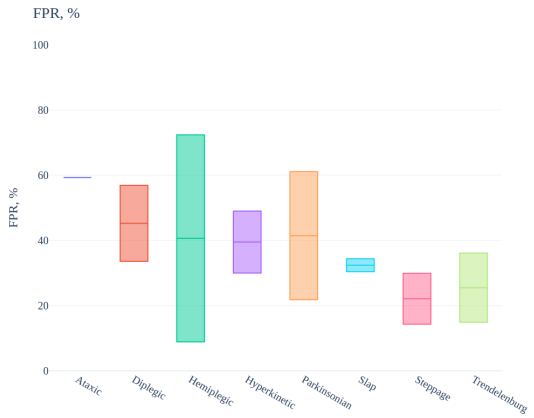

Fig. 4: Distribution of False Positive Rate for all persons for different gait types. On y-axis is False Positive Rate in percents, on x-axis is different anomalies. Most frequent results are in the boxes, and outliers are shown by whiskers and dots.

recordings of second person low true positive rate (Fig. 3) could be observed: 70%, 68%, 36%, 50%, 67% and 71% respectively. For other gait types true positive rate was above 83%, where best results were for Ataxic and Hemiplegic gait types for first person and Slap gait type for second person with F1 score of 100%. Average true positive rate remains high.

On the other hand false positive rate (Fig. 4) was high for most of the gait types, excluding Parkinsonian gait type for first person and Hemiplegic, Steppage and Trendelenburg gait types for second person with 22%, 9%, 14% and 15% respectively, which were the lowest false positive rates in current results. Highest false positive rates was for Diplegic and Parkinsonian gait recordings for second person with 57% and 61% respectively. For first person it was high for Hemiplegic, Hyperkinetic and Ataxic gait types with 72%, 49% and 59% respectively. For all other gait types false positive rate stayed between 30-40%. High average false positive rate is main reason for low F1 scores, which shows that two class classification algorithm can struggle with such task. This is because abnormal steps can vary in shape, which can confuse such algorithm and lead to misclassification.

For earliness measure (Fig. 5) it could be observed, that for most gait types earliness is less than one second. Typical step duration is usually around 1.2-1.4 seconds for persons in this paper. For Steppage gait type most common earliness measure is around 1-1.2 seconds, which is end of a step. However for Steppage gait faster detection is needed, because anomaly is happening at the beginning of a step, during swing phase. For other gait types it could be observed, that detection was mainly in the middle of a step, which shows, that algorithm can detect anomaly early, during the swing phase of a step.

As can be seen, RTtsSVM-AD algorithm is capable to detect abnormalities in a human gait in real-time as achieved earliness results showed. These detection results are corresponding to the moments of anomaly occurrence in chosen gait types. For

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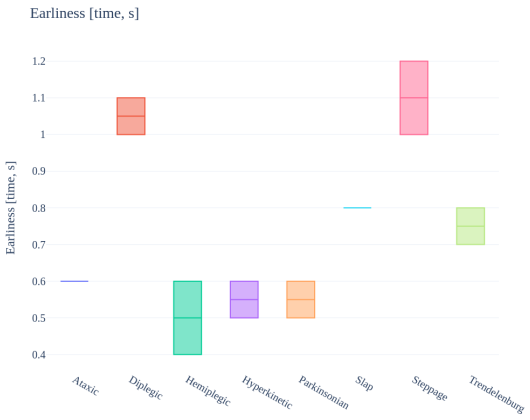

Fig. 5: Distribution of Earliness for all persons for different gait types. On y-axis is earliness in seconds, on x-axis is different anomalies. Most frequent results are in the boxes, and outliers are shown by whiskers and dots.

most of the mentioned gait types main anomaly is happening in forefoot motion, which is captured by IMU placed on forefoot. Gyroscope angular velocities from this sensor were used in this work. Even with small training dataset, this algorithm was able to detect abnormalities in real-time.

This research have following limitations: simulated gait could have differences from actual patients gait. However it should not be an issue for the performance of the proposed algorithm. Data collected from two persons is enough to prove the concept, that real-time anomaly detection is possible, however for conclusion more data should be collected.

V. Conclusion

Simulated gait deviations data was collected by two healthy volunteers. RTtsSVM-AD algorithm was developed, by introducing preprocessing and post-processing of streaming data, to be able to use it with classical machine learning algorithm like *tslearn SVM*.

Real-time classification results, where classification is performed during the ongoing step, are different from regular classification results, where classification is performed after the full step. Thus, they can not be compared directly. This is why proposed algorithm could be used as a base method, due to lack of such algorithms for real-time in-step gait abnormality detection.

Best results were observed for Steppage, Parkinsonian and Ataxic gait types with F1 scores of 57%, 53% and 52% for first person respectively and for Slap, Steppage and Hemiplegic gait types with F1 scores of 65%, 58% and 50% for second person respectively.

In future works it is planned to test presented and other algorithms, as well as to collect additional data from more persons, with different age, weight and height.

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